

DSCC FINAL PROJECT SPARK PERFORMANCE ANALYSIS:

Pipeline Performance Overview:

The Databricks pipeline successfully executed an end-to-end tweet sentiment workflow consisting of Bronze, Silver, Gold, and aggregation layers. The automated job completed successfully in approximately 3 minutes and 45 seconds and refreshed the dashboard after pipeline completion. The pipeline processed tweet ingestion, text cleaning, sentiment prediction, and aggregation for dashboard visualization.

1. Shuffle Analysis

Aggregation by mention causes shuffle because Spark must group rows by mention across partitions.

Recommendation: repartition by mention before aggregation and reduce unnecessary columns before groupBy. These optimizations would reduce network traffic and improve aggregation speed.

2. Data Skew Analysis

Some users/mentions appear much more frequently, such as @mileycyrus and @ddlovato. This can create skew partitions during aggregation e.g groupBy. Skew causes certain Spark tasks to process much larger partitions, increasing execution time and reducing cluster efficiency.

Recommendation: monitor top keys, use salting for very large datasets, and pre-aggregate per partition.

3. Serialization and UDF Overhead

The Gold layer was slow because MLflow Spark UDF loads the transformer model and serializes data between Spark and Python workers. This additional processing can increase runtime and memory usage.

Recommendation: batch inference, use smaller model, cache cleaned_text, or run inference once and store predictions.

4. Storage Optimization

The pipeline stores processed data as Delta tables in Unity Catalog. As streaming datasets grow, Delta tables may generate many small files, reducing query performance. **Recommendation:** Delta Lake optimization techniques such as “optimize” and “zorder” on common query columns such as mention and timestamp could improve read efficiency and reduce storage fragmentation.

Expected Impact:

These optimizations reduce shuffle cost, lower task runtime, improve dashboard query speed, and make the pipeline more scalable.

Jobs & Pipelines >

 Final_Project_Jobs

  Send feedback



Run now

RunsTasks

Run total duration

May 06

3m 45s

1m 52s

Tasks

Run_Tweet_Pipeline

Refresh_Dashboard

View latest successful run

Cancel runs

Start time	Run ID	Laun...	Durat...	Sta...	Error ...	Run ...
May 06, ...	39471...	Manual ly	3m 46s	✓ S.		

Schedules & Triggers

At 02:00 AM (UTC+00:00 — UTC)

Edit trigger

Pause

Delete

Job parameters

No job parameters are defined for this job

Edit parameters

Tags

Add tag

Job notifications

mwaibanjehop...

On failure

Jobs & Pipelines > Final_Project_Jobs > Run 394715074045282 >

 Run_Tweet_Pipeline run

 Succeeded

Edit task

Repair task

Final_project_pipeline

Open pipeline details

tweets_bronze

tweets_silver

tweets_gold

gold_tweet_a

Details

Job ID

1064160720287932

Job run ID

394715074045282

Task run ID

16862889513907

Run as

mwaibanjehope@gmail.com

Status

Completed

Tables 4Performance 4

Filter by table

View all in query history

Statement	Started At	Duration	Rows read	Bytes read	Bytes written
> REFRESH MATERIALIZED VIEW workspace.default.gold_tweet...	May 06, 2026, 12:32 AM	2 s 327 ms	0	0 B	0 B
> REFRESH STREAMING TABLE workspace.default.tweets_gold ...	May 06, 2026, 12:32 AM	1 s 587 ms	0	0 B	0 B
> REFRESH STREAMING TABLE workspace.default.tweets_silve...	May 06, 2026, 12:32 AM	2 s 335 ms	0	0 B	0 B
> REFRESH STREAMING TABLE workspace.default.tweets_bronz...	May 06, 2026, 12:32 AM	5 s 271 ms	0	0 B	0 B

May 06, 2026, 12:30 AM

2m 36s

Refresh all